PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Implement Client Script & UI Policy (Incident)

DATE	TEAM ID	PLATFORM	MAX MARKS
01.10.2026	SWTID-2026-6297	ServiceNow	4 Marks

Lead: Anirudh **Members:** Shyam Sundar, Mohan Raj

Technical Architecture



Architectural Guidelines & Demarcation:

- **Application Logic:** Enforces dynamic field behavior, mandatory status, visibility, and save-time validation on Incident forms.
- **Infrastructural Demarcation:** Native ServiceNow SaaS Cloud platform deployment. Client browser processes local validation rules asynchronously.
- **Interfaces & Data Storage:** Communicates with standard ServiceNow Cloud DB tables (`incident`, `sys_user`) via GlideAjax and REST APIs.

Table 1: Technical Components & Architecture Stack

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
1	**User Interface**	ServiceNow Incident Form Views & Service Operations Workspace.	**HTML5, CSS3, JavaScript, GlideForm API**
2	**Application Logic-1**	UI Policy Engine for dynamic field visibility, read-only status, and mandatory rules.	**ServiceNow UI Policies & Policy Actions**
3	**Application Logic-2**	Client-side Scripts executing onChange, onLoad, and onSubmit validation logic.	**ServiceNow Client Scripts (GlideForm / g_form)**
4	**Application Logic-3**	Asynchronous data fetching for caller profiles and configuration items without form freeze.	**GlideAjax & Script Includes (JavaScript)**

Table 1: Technical Components (Continued)

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
5	**Database**	Relational database storing Incident records, audit history, and configuration items.	**ServiceNow Relational DB Engine (incident table)**
6	**Cloud Database**	High-availability cloud metadata storage for client script definitions and system rules.	**ServiceNow Enterprise Cloud Storage**
7	**File Storage**	Attachment storage for screenshots, logs, and evidence attached to Incidents.	**ServiceNow Cloud File & Attachment System**
8	**External API-1**	ServiceNow Table API for third-party monitoring tools creating automated incidents.	**ServiceNow Table API / REST Services**
9	**Infrastructure**	Enterprise SaaS multi-instance cloud deployment with automated failover.	**ServiceNow SaaS Enterprise Cloud**

Table 2: Application Characteristics

S.NO	CHARACTERISTICS	DESCRIPTION	TECHNOLOGY
1	**Open-Source Frameworks**	Standard ECMAScript execution engine supporting client-side scripts.	**JavaScript (ES6+ Standard)**
2	**Security Implementations**	Role-based access control (RBAC) and Access Control Lists (ACLs) preventing unauthorized field edits.	**ServiceNow ACLs, User Roles (itil, admin), TLS 1.3**
3	**Scalable Architecture**	Client-side execution reduces server load, allowing seamless scaling for high concurrent user volumes.	**Multi-Instance Architecture, Client-Side Execution**
4	**Availability**	High availability with redundant cloud infrastructure ensuring continuous platform access.	**ServiceNow Advanced High Availability (AHA)**
5	**Performance**	Client scripts optimized to minimize synchronous server calls and optimize form load time.	**Asynchronous GlideAjax, DOM-Free Scripting**

References:

• ServiceNow Documentation: *Client Scripts & UI Policies Architecture Guide*
• C4 Model Architectural Framework: *https://c4model.com/*
• ServiceNow Developer Docs: *GlideForm (g_form) & GlideAjax API References*